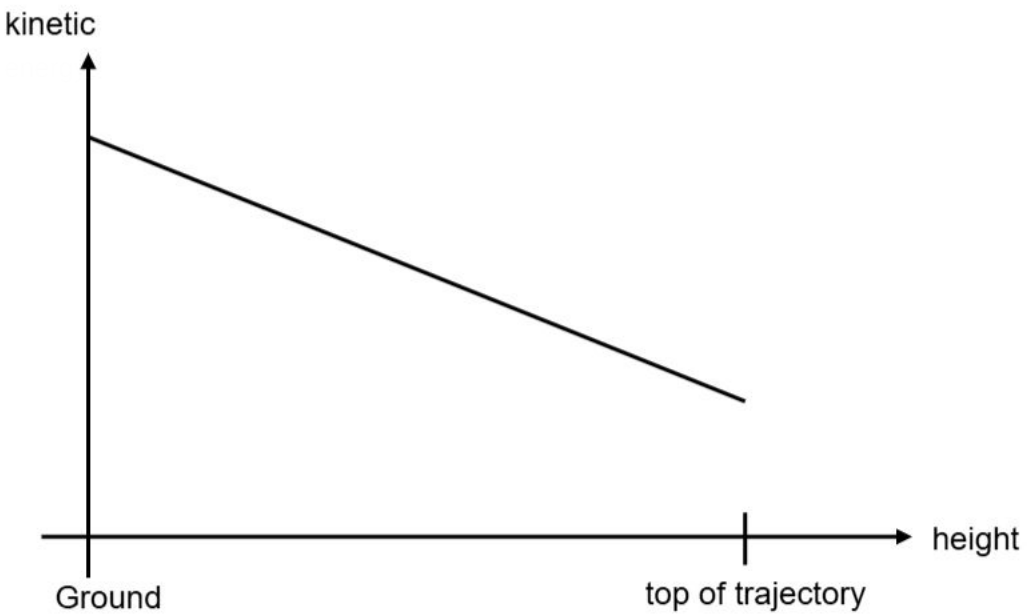
| Question 2 | | |
|---|---|---|
| **(a)(i)** | Vertical velocity = horizonal velocity when the resultant velocity is 45° to the horizontal.<br>$v_y = v_x = 25\cos(53°) = 15.045 = 15 \text{ m s}^{-1}$ | **A1** |
| **(a)(ii)** | Taking upward as positive,<br>$v_y = u_y + a_y t$<br>$15 = 25\sin(53°) + (-9.81)t$<br>$t = 0.50$ s | **A1** |
| **(b)** | $KE_{8.0} = KE_{ground} - GPE_{8.0}$<br>$KE_{8.0} = \frac{1}{2}(0.20)(25)^2 - (0.20)(9.81)(8.0)$<br>$= 47$ J<br>Alternative: Accept Kinematics method. | **M1**<br>**A1** |
| **(c)** |  <br>**B1** – straight line with decreasing KE<br>**B1** – non-zero KE at top of trajectory | |